

Econ 306 Summer Session I

Quiz 2

You may not use any class materials or AI tools to answer the question. Write your answer on a separate piece of paper and then take a photo/scan your paper and upload to ELMS at the end of the quiz. You will have 20 minutes to complete the quiz and then 10 minutes to upload your answers. Round any decimal answers to 1 decimal place or leave your answers in fractional form (For example $\frac{50}{3}$). The quiz is out of 9 points. Please show your work as partial credit may be given for correct work. You are welcome to use a calculator.

Perfect Competition

Consider a perfectly competitive market where all firms have identical cost functions given by:

$$C(q) = 2q^2 + 20q + 200$$

The market demand is:

$$Q_D = 400 - 5p$$

1. What is the shut-down price for each firm? (3 points)
2. If there are currently 5 firms in the industry, what will be the short-run equilibrium price and market quantity? (3 points)
3. What will be the long-run equilibrium price, market quantity, and number of firms? (3 points)

Cost Functions (long run and short run):

$$C(Q)$$

Total Cost

$$AC(Q)=C(Q)/Q$$

Average Cost

$$MC(Q)=C'(Q)$$

Marginal Cost

Short-Run Cost Curves:

$$C(Q)=VC(Q)+FC$$

Total Cost

$$VC(Q)$$

Variable Cost

$$FC$$

Fixed Cost

$$AVC(Q)=VC(Q)/Q$$

Average Variable Cost

$$AFC(Q)=FC/Q$$

Average Fixed Cost